Assignment

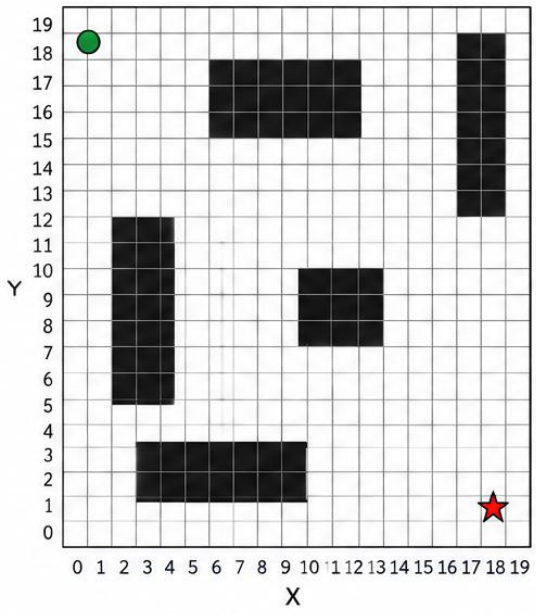
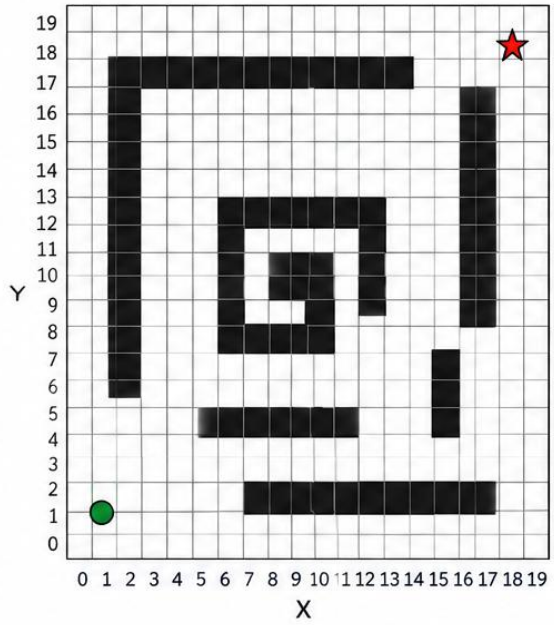
The students have to implement and compare their respective evolutionary algorithms and heuristic algorithms in corresponding environments which is mentioned in the tabulations Table 3. Evaluate path length, computation time, success rate, number of turns, and collision-free navigation.

Table 1

S. No.	Algorithm	Category	Why use it for path planning?
E1	Genetic Algorithm (GA)	Evolutionary	Evolves candidate paths using selection, crossover and mutation.
E2	Particle Swarm Optimization (PSO)	Swarm intelligence	Refines candidate paths using individual and group search experience.
E3	Ant Colony Optimization (ACO)	Swarm intelligence	Builds paths using pheromone trails and heuristic desirability.
E4	Differential Evolution (DE)	Evolutionary	Uses population differences to explore and improve candidate solutions.
E5	Grey Wolf Optimizer (GWO)	Nature-inspired metaheuristic	Balances exploration and exploitation using a leadership hierarchy.
E6	Artificial Bee Colony (ABC)	Swarm intelligence	Explores paths through employed, onlooker and scout bee search strategies.
E7	Whale Optimization Algorithm (WOA)	Nature-inspired metaheuristic	Uses encircling and search strategies to improve candidate solutions.
E8	Simulated Annealing (SA)	Probabilistic heuristic	Sometimes accepts worse candidate paths to escape local optima.
E9	Cuckoo Search (CS)	Nature-inspired metaheuristic	Uses Lévy-flight-inspired exploration and replacement of weaker solutions.
E10	Firefly Algorithm (FA)	Swarm intelligence	Moves candidate solutions toward more attractive, higher-quality solutions.

Table 2

S. No.	Algorithm	Abbreviation	Main principle
H1	A* Search	A*	Combines the cost travelled \(g(n)\) and estimated remaining cost \(h(n)\).
H2	Greedy Best-First Search	GBFS	Expands the node with the lowest heuristic distance to the goal.
H3	Weighted A*	WA*	Gives additional weight to the heuristic to guide the search more aggressively.
H4	Jump Point Search	JPS	Skips symmetric intermediate nodes on uniform-cost grids.
H5	Theta*	Theta*	Uses line-of-sight checks to find paths with fewer grid-direction constraints.
H6	D* Lite	D* Lite	Reuses previous search information when the environment changes.
H7	Lifelong Planning A*	LPA*	Reuses search results when graph costs change.
H8	Beam Search	BS	Keeps only a limited number of promising nodes at each search level.
H9	Bidirectional A*	Bi-A*	Searches from the start and goal, aiming to connect the two searches.
H10	Fringe Search	FS	Uses iterative cost thresholds as an alternative to A*'s priority queue.

Map No	Layout
1	
2	
3	